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Personal Protective Equipment Detection for Construction Workers: A Novel Dataset and Enhanced YOLOv5 Approach

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ABSTRACT Current research on personal protective equipment (PPE) detection has mainly focused on hard hats, overlooking the detection of reflective clothing. Therefore, this study aims to address this research gap comprehensively. We achieve this by creating a novel dataset using semi-automatic labeling techniques and enhancing the YOLOv5 model. The dataset consists of four categories, assessing the presence of hard hats and reflective clothing. Additionally, we introduce an attention mechanism and an improved loss function to tackle challenges related to detecting reflective clothing and overlapping detection frames. Through extensive multi-model comparison experiments, our improved model, AL-YOLOv5, outperforms the baseline model with remarkable advancements of 0.9 AP in the data-limited category and 0.4 mAP overall. Notably, our improved model shows substantial progress in detecting reflective clothing, significantly reducing false detections, and improving overlapping bounding frames. In conclusion, this study contributes to PPE detection accuracy through a novel dataset and an improved model.

INDEX TERMS Attention mechanisms, new dataset, personal protective equipment (PPE), reflective clothing, SEIoU

I. INTRODUCTION

The construction site in the construction industry relies on a large number of construction workers to work manually, and the complex construction environment brings great risks to workers[1-3]. Thus, how to effectively manage complex and mobile workers is a challenge to the current construction industry[4]. Although the managers of construction sites require workers to wear appropriate personal protective equipment (PPE) to reduce or even avoid injuries of workers caused by possible accidents on construction sites[5], due to lack of safety awareness, inconvenient wearing, and other reasons, workers often neglect to wear hard hats and reflective clothing in actual construction, resulting in personal injuries on construction site in the event of safety accidents[6, 7]. Meanwhile, due to the lack of automatic monitoring and management, the current management is still based on manual supervision, which is inefficient. The neglect of safety problems by both workers and managers leads to the common phenomenon of not wearing safety protection equipment on construction sites, thus resulting in huge potential safety hazards[8].

As an emerging intelligent safety management technology[9, 10], Computer vision technology enables the safety management of PPE detection for construction workers without the need for wearable or touch-based sensors directly on the workers[11]. Instead, it relies on existing non-intrusive sensors, such as cameras, to analyze images, thereby enhancing worker management and improving safety during construction processes[12-14]. As such, it has aroused in-depth discussion in academic circles. At present, the most common and important PPE on construction sites are hard hats and reflective clothing, but most studies based on computer vision only focus on detecting whether workers wear hard hats[15-18], and ignore the importance of reflective clothing. It is important to know that although reflective clothing will not directly protect workers' bodies, it can highlight workers' locations to a great extent, avoid collision between machinery and workers, and reduce the risk of collision accidents[3]. Moreover, the variability and overlapping of the features of reflective clothing lead to difficulty in the detection of reflective clothing, which is obviously different from the detection of hard hats. Therefore, the research on automatic detection of reflective clothing is also extremely important. Furthermore, there are some problems in many studies on PPE detection at present, such as insufficient sample data, unbalanced

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datasets, lack of real construction scene information, etc., which result in unsatisfactory generalization and detection precision of PPE detection[19-22]. There, detecting PPE in construction scenes, particularly reflective clothing and hard hats, is of utmost importance for ensuring worker safety.

Reflective clothing presents unique challenges due to its susceptibility to obscurity and deformation caused by factors like worker posture, object occlusion, and the surrounding environment. In contrast, hard hats exhibit relatively more fixed and uniform features. Therefore, addressing the automatic detection of reflective clothing becomes a critical aspect. Additionally, the task of target detection encounters frequent overlapping detection frames, requiring an effective bounding box regression loss function to enhance accuracy. The commonly used YOLOv5 method employs the CIoU loss function, which considers the bounding box's aspect ratio as a penalty term, but still has some ambiguity in solving reflective clothing feature deformation and overlapping. Moreover, incorporating the attention mechanism is becoming increasingly important to enable selective focus on essential regions in the detection process [23-25]. Introducing the attention mechanism can help prioritize key features of PPE, such as reflective clothing and hard hats, thereby enhancing detection accuracy and addressing the issue of overlapping detection frames.

In order to effectively solve the current problem of ignoring reflective clothing for protecting workers' safety in PPE detection, as well as the inspection interference problem caused by the variation of features in detecting different types of PPE devices, we propose to construct a comprehensive dataset to include the detection of whether workers are wearing reflective clothing and hard hat. We propose to construct a comprehensive dataset to include the detection of whether workers are wearing reflective clothing and hard hats, introduce an attention mechanism to solve the feature overlap problem caused by PPE detection, and adopt an improved loss function to enhance the accuracy of the detector in recognizing feature changes in PPE images. By implementing these measures, we aim to effectively address the current problems of sample imbalance, detection frame overlap, and accuracy in PPE detection. Ultimately, our approach seeks to provide intelligent decision support and improve safety management in construction scenarios.

II. Literature Review

A. Study on PPE Data Set

It is well known that a dataset is the basis of object detection. At present, the research on workers' PPE datasets focuses more on hard hats, and their lack of research on reflective clothes. Iannizzotto (2021)[26] and Nath (2020)[8] enhanced PPE detection but faced challenges in achieving precision for reflective clothing, with detection rates as low as 60% and 70%, respectively.

A large number of data reports and studies have pointed out that the visibility of construction sites is extremely important for workers' safety. Especially in the construction of bridges and tunnels and the frequent night construction, wearing reflective clothing to ensure workers' safety and visibility can effectively avoid collisions between workers and machinery[3, 27]. Reflective clothing, though essential, poses detection challenges due to its deformable features and interference with regular clothing. Its importance, equivalent to that of a hard hat, necessitates the use of deep learning models for automatic detection.

In addition, the existing research mainly focuses on the analysis of different model frameworks used for the classification of PPE status and ignores the data set structure, that is, the problem of category imbalance. It may significantly reduce the accuracy of some categories in PPE detection[28]. Cheng[28] and Nath, while exploring reflective clothing detection, treated it similarly to hard hats, neglecting its deformable nature. This results in limited effectiveness, especially as workers are less accepting of reflective clothing, causing imbalances in training samples.

Considering the absence of uniform datasets like PASCAL VOC[29] and COCO[30] for construction sites, scholars have constructed new datasets[31, 32], for detecting construction workers and equipment. However, in terms of the PPE identification task, these datasets face challenges. Wu et al. (2019)[19] achieved an 83.89% (mAP) precision but was limited by data volume. Nath's YOLOv3-based approach achieved real-time processing, but 1,500 labeled images and about 4,700 examples of training data led to a lower precision of 72.3% (mAP). Mneymneh (2019)[20] detected whether workers wore hard hats in different scenes and situations with only a few test images, the situations where hard hats were blocked and there were various kinds of hard hats were not considered. so the confidence of the results needs to be further evaluated based on more comprehensive data sets.

Despite efforts to collect construction site images for object detection[19, 33-35], existing datasets often lack diversity in representing different construction scenarios. Studies like Li et al. (2022)[36] face limitations due to data source discrepancies and a lack of real-world construction site complexities, impacting the effectiveness of the results based on this model in the real scene. The special environment of the construction site leads to different conditions of its images. The different conditions, such as illumination, image shooting angle, and shooting distance,

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will all affect the detection precision[37]. Therefore, in the model training, it is better for the data set to acquire real images from the construction site and contain all real situations on the spot as much as possible.

In conclusion, the current PPE data set is unbalanced and divorced from reality, with insufficient data and difficulty in manual labeling, which may be the reasons that limit the construction of a balanced data set. In addition, the detection of reflective clothing is important and necessary, so it is necessary to expand the categories of PPE data sets. Therefore, it is required to construct a brand-new data set with a faster and more accurate method to incorporate various situations on the construction site for more accurate detection of hard hats and reflective clothing.

B. PPE Detection Technology Based on Deep Learning

Efficient automatic management of workers' safety gear, facilitated by technologies like intelligent construction sites and equipment, is paramount for contemporary safety management[9, 38, 39]. Current computer-based PPE detection technologies can be broadly categorized into sensor-based and vision-based technologies.

Sensor-based methods involve sensor installation for signal analysis, utilizing sensing technologies like GPS and RFID[17, 40]. While effective, these methods incur significant costs and management burdens. For instance, Kelm et al. [41] proposed a passive RFID tag for PPE compliance, yet the reconstructed PPE and binding identity information with PPE bring a huge cost and management burden. Similarly, hard hat monitoring systems[17] based on real-time positioning are costly and pose challenges to worker acceptance.

Vision-based technology, on the other hand, relies on cameras to supervise workplaces. It will not interfere with the workers' construction process, and no additional equipment is needed. It is only required to read the information in data images and videos through a computer, and then manage the site through analysis[42, 43]. Object detection through computer vision (CV) in construction sites primarily focuses on monitoring workers, identifying unsafe behaviors, and ensuring PPE compliance[36, 44-46]. Detection methods are broadly categorized into two-stage and single-stage methods[47].

Two-stage methods, employing region proposals and convolutional neural networks (CNN), such as Faster R-CNN and ResNet-50, offer high precision but struggle with real-time monitoring. Fang et al.(2018) [37] checked whether workers wore hard hats by the Faster R-CNN method based on a great number of construction site images. For real-time monitoring of construction sites, it cannot achieve quite a good effect, and the average speed in different scenes is about 5 frames/S. The previous studies have defined "real-time" as processing at least 5 video frames per second(FPS) [48], while the current research thinks that the standards of real-time detection should be improved[47, 49]. Moreover, the real-time requirement on construction sites is relatively high. Therefore, faster safety detection can prevent workers from being in danger.

YOLO (You Only Look Once) series, including YOLOv5, provides a faster alternative by completing classification and positioning in a single pass[47], realizing end-to-end detection and enabling real-time detection of object targets in real scenarios[50], making them well-suited for real-time construction site monitoring. Particularly, YOLOv5's speed and precision make it suitable for accurate PPE detection, including hard hats and reflective clothing[51].

Higher detection accuracy in the YOLOv5 model can be achieved through the addition of the attention mechanism, one of the most widely used modules in computer vision [52]. Integrating an attention mechanism into the YOLOv5 model enhances feature extraction and improves detection accuracy[53, 54]. For instance, Chen et al. (2023) [55] integrated the Convolutional Block Attention Module (CBAM) for real-time detection of strawberry diseases. In complex construction environments, the attention mechanism proves crucial, highlighting key features and suppressing irrelevant information. Similarly, Liu et al. [53] introduced the Coordinated Attention (CA) module, which extracts features from detected objects and suppresses irrelevant features, further improving detection accuracy.

Adding the attention mechanism is particularly important in complex construction environments, where reflective clothing and hard hats may blend with the background, making accurate inspection challenging. Moreover, the similarity of features between opposing categories can make it difficult to accurately distinguish PPE. The introduction of the attention mechanism allows the model to better focus on accurately detecting the categories of interest, providing a viable solution to the problem of overlapping reflective clothing and hard hats in complex environmental backgrounds.

To enhance detection accuracy in YOLOv5, the loss function plays a pivotal role, in refining the distance between predicted and real bounding boxes through bounding box regression (BBR), classification, and target loss. The conventional IoU measure faces challenges when boxes don't overlap, especially prevalent in PPE detection due to varying geometries, the different geometries of the hard hat and reflective clothing labeled images make it difficult to accurately obtain the correct regression of the bounding box based on the overlap area loss function.

Numerous adaptations like GIoU[56], CIoU[57], EIoU[58], Alpha IoU[59], and SIoU[60] introduce penalty terms to address these issues and improve convergence. The BBR has three key problems of regression loss: the overlap area of the predicted boxes and the target boxes P1, the distance of the center point P2, and the aspect ratio of the height and width P3, which can be written uniformly as $L_{loss} = 1 - IoU + P_i$. While YOLOv5 employs CIoU, it struggles with aspect ratio limitations. Scholars propose EIoU, converting aspect ratio (P3) into height and width, minimizing disparities between target and anchor boxes, and boosting effectiveness. Gevorgyan(2022) [60] further

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considered the vector angle between the target boxes and the predicted boxes, proposed a new loss function SIoU, and constructed a new penalty term including Angle cost, Distance cost, and Shape cost. While improvements in individual aspects show enhanced detection, the variability and overlap in PPE bounding boxes demand heightened accuracy. Thus, a new loss function can be constructed by amalgamating the P3 enhancement from EIoU with the P2 refinement from SIoU, aiming to boost detection accuracy and efficiency in object detection (OD).

In summary, there are currently three major problems with the identification of PPE for construction workers:

a. Limited analysis of reflective clothing: Currently, PPE detection focuses primarily on hard hats, overlooking the critical aspect of reflective clothing, which plays an equally important role in ensuring worker safety. There is a lack of a new dataset that includes both hard hats and reflective clothing, which hinders comprehensive PPE detection research.

b. Imbalanced and inefficient image data: The current PPE image dataset suffers from imbalance and detachment from real-world construction scenarios. Different poses and category characteristics of workers during construction create variations that limit the generalization and effectiveness of the dataset. In addition, the scarcity of data and the challenges of manual labeling hinder the creation of a balanced dataset.

c. Challenges in real-world design scenarios: Reflective clothing often faces problems of obscuration and deformation due to worker poses, object occlusion, and the environment. These deformations and occlusions significantly increase the difficulty of detection, necessitating model improvements and the adoption of faster and more accurate detection methods for PPE research.

To address these, this paper makes significant contributions:

1. **Construction of a novel PPE dataset:** Our dataset includes two common construction scenarios: workers wearing and not wearing hard hats and reflective clothes. This particular aspect introduces interference between the four classes. In order to simplify the difficult task of annotating the dataset, we used semi-automatic annotation techniques. This approach increases the efficiency of the labelling process.
2. **Attention mechanism and improved SEIoU loss function:** An attention mechanism integrated into YOLOv5 and an improved SEIoU loss function enhance the model's ability to detect reflective clothing and hard hats accurately. This addresses challenges of feature overlap and deformation in PPE detection.".

III. Methodology

The OD process based on the CV is generally similar. The PPE detection process in this paper was mainly divided into the following steps: data collection and processing, model building, model training, and testing, automatic labeling, construction of a complete dataset, and final training and detection. The concrete implementation framework is shown in Fig.1.

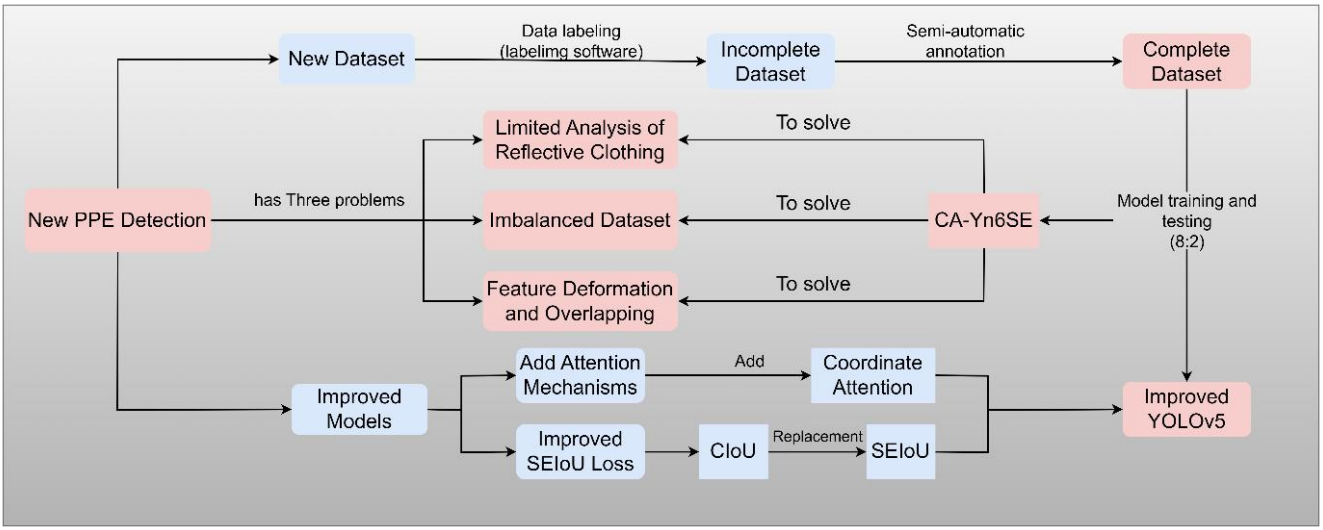


FIGURE 1. PPE Detection Process Based on New DataSet and Improved YOLOv5

A. Dataset Creation and Processing

In the present study, a brand-new dataset of PPE detection for construction workers was built. Most of the images in the dataset come from real construction scene shooting, and video streams and picture streams of some real construction sites acquired from the network, mainly including four categories, namely, wearing hard hats (Hat), not wearing hard hats (Person), wearing reflective clothing (Reflective-clothes), and

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wearing other clothes (Other-clothes). Please refer to Table 1 for category descriptions.

TABLE 1
DESCRIPTION OF FOUR DETECTION CATEGORIES

Categories of the wearing of PPE by workers	Label categories in the image	Description
Wearing hard hats	Hat	The category that should be worn normally
Not wearing hard hats	Person	Opposite sample of Hat
Wearing reflective clothing	Reflective-clothes	The category that should be worn normally
Wearing other clothes	Other-clothes	Opposite sample of Reflective-clothes

The dataset basically included the situation that the hard hats and reflective clothes are blocked or deformed with the human body, which could truly reflect the wearing condition of hard hats and reflective clothing on the construction site. In addition, background images of 1% were also added to the dataset to improve the mAP of object detection. The corresponding reason is that the characteristics of the construction site determine that it is usually disorganized and full of obstructions. At the same time, it is difficult to obtain the best distance and viewpoint when taking pictures at the construction site, and the objects in the images may be extremely small, or partially blocked and cluttered. Therefore, to keep the dataset consistent with the construction site, images in the dataset should be acquired from the construction site to simulate its actual situation. Moreover, to ensure the balance in the number of labels of the 4 detection categories, some images of not wearing hard hats were added. The image dataset is shown in Fig.2.

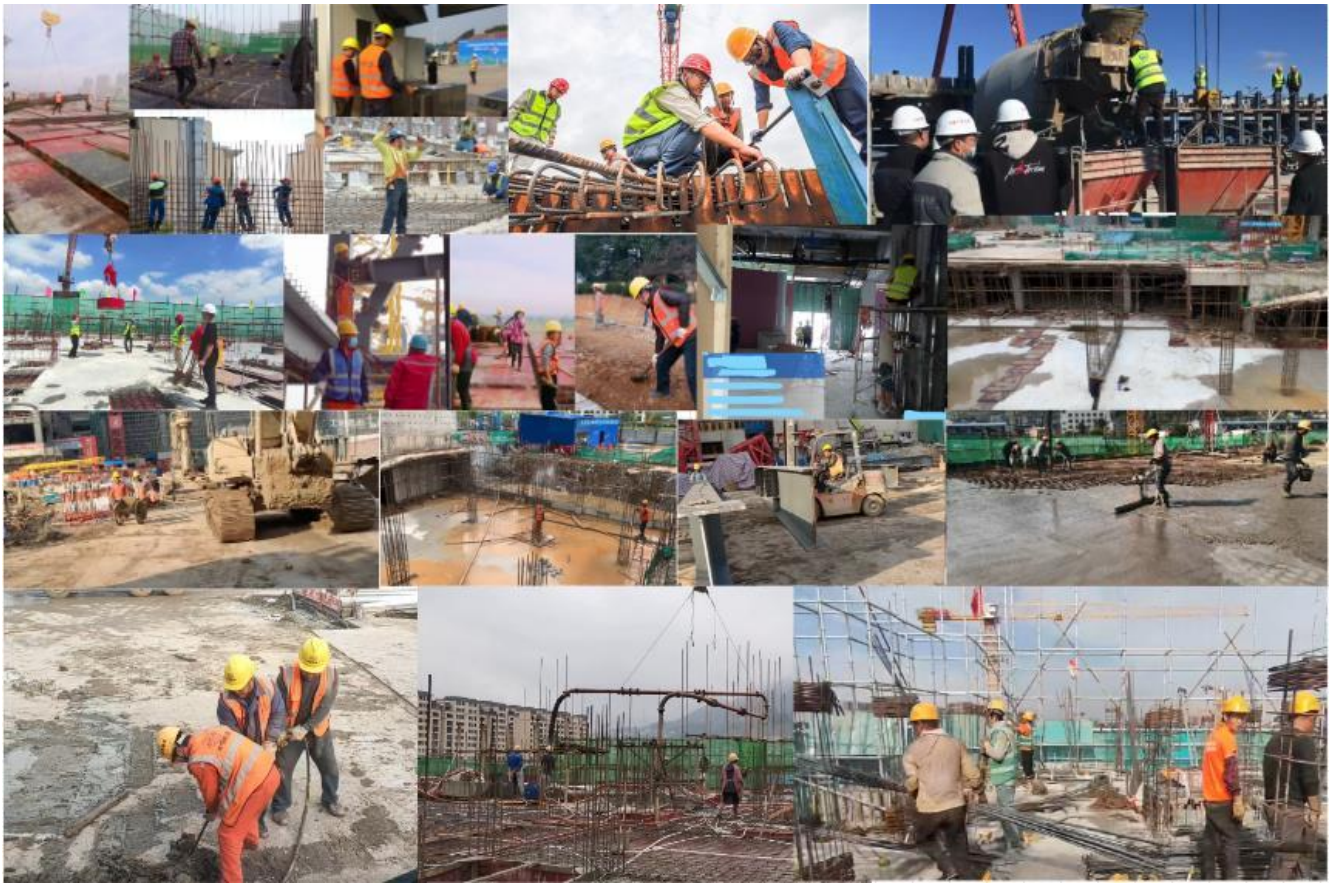


FIGURE 2. Display of Construction Site Image Dataset (Partial)

In this paper, the field personnel under PPE management research should involve all the people who enter the construction site. Therefore, workers herein generally refer to all persons who enter the construction site. After the relevant image and video datasets are collected well, it is required to label the images to meet the input requirements of the model. It is worth noting that the labeling of the dataset in this paper can be divided into two parts, first is the annotation of the unsafe dataset, due to the inconsistency of the number of object categories in the image, the image dataset collected through the actual and network, it is difficult to ensure the balance of each category, especially the data of the detection PPE of the construction site, manually annotate the incomplete dataset, and then construct the complete dataset according to the semi-automatic annotation. The labelImg is employed for

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the labeling of the dataset. The specific operation process is as Fig 3:

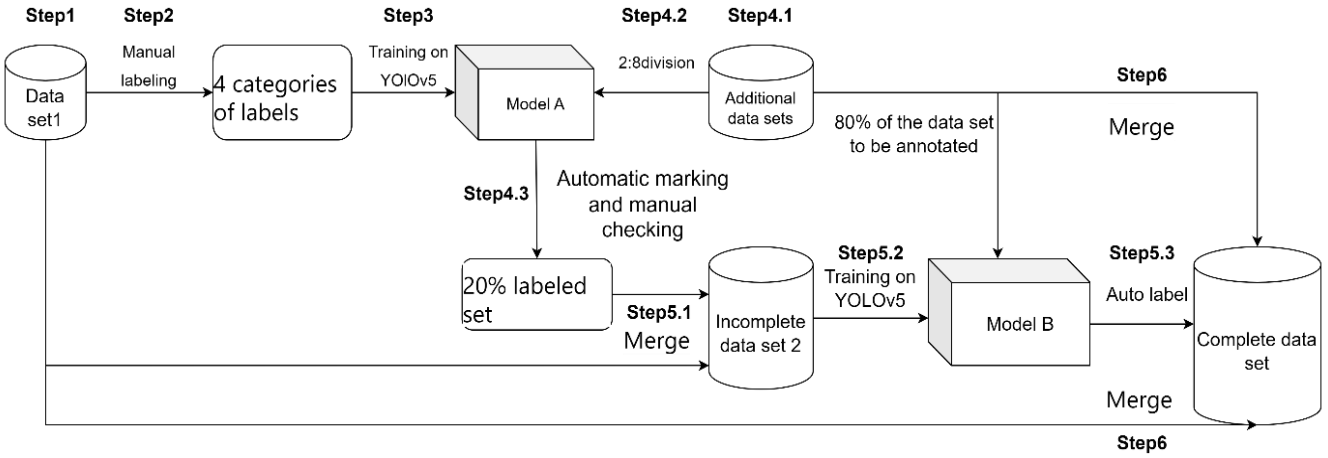


FIGURE 3. **Dataset annotation process**

Step 1: Initial Data Collection
Collect image data for hard hats and reflective clothing. Note that the current dataset mainly contains hard hat images, leading to an imbalanced dataset with fewer reflective clothing labels, referred to as incomplete dataset A.
Step 2: Manual Labeling
Manually label the initial dataset with four categories: hat, person, reflective-clothes, and other-clothes, using the txt format for labeling.
Step 3: Model A Training
Train model A on the incomplete dataset using the improved YOLOv5. However, model A exhibits unbalanced detection accuracy for each category.
Step 4: Additional Data Collection and Labeling
(1) Gather additional image categories, such as workers wearing other clothes or not wearing hard hats, to address low accuracy issues.
(2) Split the extra data stream into labeled and to-be-labeled sets with a 2:8 ratio. Input 20% of the to-be-labeled set into model A for detection.
(3) Perform YOLOv5 detection to obtain a txt format file with 4 labeled categories. Review and correct any inaccuracies using labelImg to complete the correct labeling of the 20% labeled set.
Step 5: Model B Training
(1) Combine accurately labeled image streams with the initial dataset to create dataset 2.
(2) Train model B on dataset 2 using improved YOLOv5, resulting in higher detection accuracy for the four categories and more accurate labeling for the additional dataset.
(3) Automatically label 80% of unlabeled image streams in the additional dataset using model B, ensuring the labeling aligns with correct results. However, manual review is still necessary for complete accuracy.
Step 6: Iterative Data Enhancement
Repeat steps (4) and (5) to continuously extend low-precision image data and achieve automatic annotation of additional image datasets, ensuring balanced and diverse datasets in each category.

B. 3.2 Yolov5 Framework

As the representative method of the YOLO series, YOLOv5 is characterized by higher speed, higher detection precision, and smaller file size[61]. YOLOv5 is the fifth generation of the YOLO series, which is a series of OD models with different depths trained based on COCO datasets (YOLOv5, 2021)[62]. The selection of YOLOv5 was based on its widespread use and attention during the study period, providing a robust foundation for improvements tailored to our specific goals. Although it is not the latest model, it has been widely applied in the engineering field due to excellent detection speed and high precision and has an excellent performance in OD[63-66]. Next, its main framework and key structure will be introduced. Its main structure is as shown in Fig.4:

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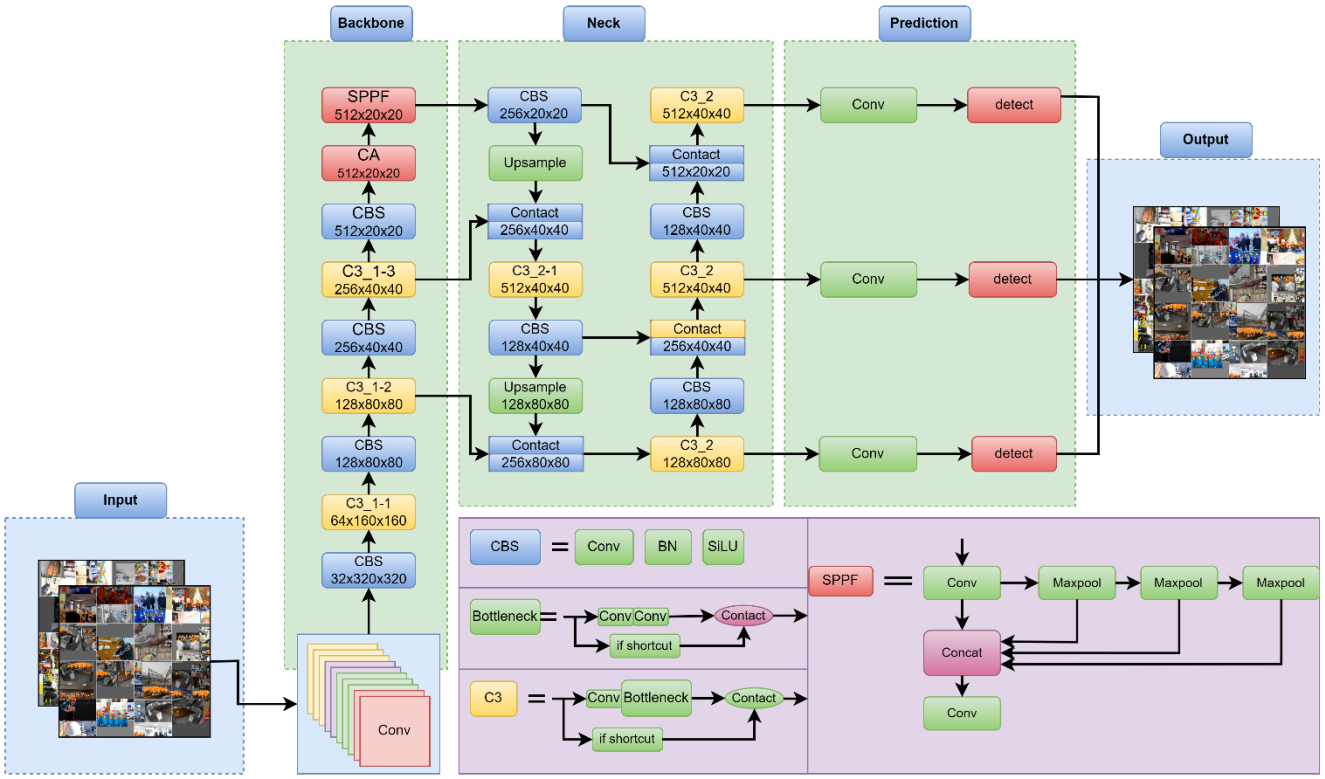


FIGURE 4. **Structure Diagram of CA-Yolov5-6.2**

The main framework is as follows:

Input: The size of the input image is 640*640. This stage usually includes an image preprocessing stage, that is, scaling the input image to network input size and carrying out normalization. Besides, the Mosaic data enhancement operation is adopted to improve the model training speed and network precision;

Backbone: A backbone network is usually some classifier network with excellent performance. This module is used to extract some general feature representations[67];

Neck: FPN and PAN structures are adopted, which can further enhance the diversity and robustness of features [68, 69];

Head: The head is used to output the object detection results.

The adaptive image scaling adopted by YOLOv5 determines that it is unnecessary to cut the image size uniformly when inputting the images, and the images from different shooting tools can be directly input into the model. In this way, it becomes easier to collect image data at the construction site.

Compared with YOLOv5-5.x, the network structure of YOLOv5-6.x is more downsized, specifically as follows: Conv replaces the Focus module to facilitate exporting other frameworks. SPPF module replaces SPP, which improves the speed of the backbone network by twice[70]; The repetition times of the C3 layer in the backbone are reduced from 9 to 6, and Conv is adopted to replace the Focus module, which can facilitate exporting other frameworks; Shortcut (C3 n=1 True) is introduced into the last C3 layer in the backbone.

C. 3.3 Performance Evaluation Index

Upon completion of the model training, it is required to evaluate the detection effect. IoU, mAP, recall, and precision are usually adopted to evaluate the model comprehensively.

$$Precision = \frac{TP}{TP + FP} \tag{1}$$

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

(1) and (2) denote the proportion of correct and complete searches, respectively.

$$AP = \int_0^1 P(R)dR \tag{3}$$

$$mAP = \frac{\sum_{i=1}^{n} AP_i}{n} \tag{4}$$

where n denotes the number of categories to be detected in the target inspection. The mAP is the area enclosed by Precision and Recall as the two axes after plotting. Where mAP@0.5 is the average AP of all detection categories when the IoU is 0.5.

D. 3.4 Model Enhancements:

1) LOSS FUNCTION IMPROVEMENT

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The effectiveness of detection in OD, i.e., the degree of matching between the target boxes and predicted boxes, depends on the loss function to a great extent. The current YOLOv5 series uses CIoU, and the main ones with fruitful improvements on CIoU are EIoU and SIoU, which have proposed new ideas on the penalty terms P2 and P3, respectively. In this paper, we mainly consider the improved fusion of the above effective penalty terms and propose the improved loss function SEIoU, so as to have a new improvement in checking task accuracy.

Therefore, this paper is considered to replace the loss function in YOLOv5 with SEIoU, so as to improve the positioning accuracy of the detection box as much as possible and check the wearing condition of PPE by workers in construction more accurately. The model derived from improving the loss function can be called YOLOv5-SEIoU.

The specific calculation process is as follows.

(I) First, the BBR loss function is:

$$L_{loss} = 1 - P1 + P2 + P3 \tag{5}$$

where P1=IoU and the improvement of the loss function are usually in P2 and P3.

(2) The base equation for CIoU is:

$$L_{loss} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \tag{6}$$

where $\rho^2(b, b^{gt})$ denotes the Euclidean distance between the center point of the predicted box and the real box, denotes the diagonal distance of the smallest closed rectangle of the two boxes, α is the weight function, and v is used to measure the variability of the aspect ratio. The second term in the formula represents the penalty term P1 for the overlap area IoU, the third term is the penalty term P2 for the distance, and the fourth term is the penalty term P3 for the aspect ratio of the prediction frame.

(3) The EIoU has improved on P3 for CIoU by splitting the aspect ratio：

$$L_{loss} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \frac{\rho^2(w, w^{gt})}{c_w^2} + \frac{\rho^2(h, h^{gt})}{c_h^2} \tag{7}$$

(4) SIoU has improved P2 by considering the angle variables on the basis of CIoU, and the specific calculation process is as follows.

SIoU has considered the variable—angle on the basis of CIoU, and the specific calculation process is as follows:

Angle cost: $$\Lambda = 1 - 2 * \sin^2\left(\arcsin(x)\right) - \frac{\pi}{4} \tag{8}$$

Where,

$$x = \frac{c_h}{\sigma} = \sin(\alpha),$$

$$\sigma = \sqrt{(b_{c_x}^{gt} - b_{c_x})^2 + (b_{c_y}^{gt} - b_{c_y})^2}$$

$$c_h = max(b_{c_y}^{gt}, b_{c_y}) - min(b_{c_y}^{gt}, b_{c_y})$$

Distance cost: $$\Delta = \sum_{t=x,y} \left(1 - e^{-\gamma \rho_t}\right) \tag{9}$$

Where, $\rho_x = \left(\frac{b_{c_x}^{gt} - b_{c_x}}{c_w}\right), \rho_y = \left(\frac{b_{c_y}^{gt} - b_{c_y}}{c_h}\right), \gamma = 2 - \Lambda$

Shape cost: $$\Omega = \sum_{t=w,h} \left(1 - e^{-w_t}\right)^{\theta} \tag{10}$$

Where, $w_w = \frac{\left|w - w^{gt}\right|}{max(w, w^{gt})}, w_h = \frac{\left|h - h^{gt}\right|}{max(h, h^{gt})}$

Total loss: $$L_{loss} = 1 - IoU + \frac{\Delta + \Omega}{2} \tag{11}$$

where denotes the penalty term P2 for distance loss, denotes the improvement of shape loss P3, P3 still fails to achieve the measurement of the separate longitudinal and transverse differences of the bounding box.

(5) In summary, taking the improvement points in EIoU and SIoU, while improving the penalty terms P2 and P3, the SEIoU can be written as:

$$L_{loss} = 1 - IoU + \left(\Delta + \frac{E_{P3} + \Omega}{2}\right) / 2 \tag{12}$$

where, $E_{P3} = \frac{\rho^2(w, w^{gt})}{c_w^2} + \frac{\rho^2(h, h^{gt})}{c_h^2}$) , Δ denotes the penalty term P2 for distance loss, $\frac{E_{P3} + \Omega}{2}$ denotes the penalty term P3 after considering both the distance difference based on the angle variable and the separate difference in the length and width of the bounding box.

2) ADDING ATTENTION MECHANISM

To further improve the feature extraction capability and to address the potential performance loss caused by the improvements, we incorporated an attention mechanism into the improved model. There are several common attention mechanisms, including squeezing and excitation attention (SE) [24], effective channel attention (ECA) [71], convolutional block attention module (CBAM) [25], and coordinate attention (CA) [72]. In this paper, we decided to use the CA module because of its better performance and adaptability in solving the shape deformation and occlusion problems in reflective clothing and hard hat detection tasks.

The CA module combines channel and spatial attention mechanisms, enabling it to focus on both channel and spatial information within the feature map. In PPE detection tasks, reflective clothing and hard hats often undergo shape changes due to workers' postures or object occlusions, resulting in higher variability in their features and more complex detection requirements. The CA mechanism helps the model to spatially emphasize important features in specific regions, thus improving the accuracy and robustness of target localization by allowing the model to focus more on learning region-specific details.

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One of the notable strengths of the CA module lies in its ability to gather data along two separate spatial axes and decompose channel attention into two one-dimensional coded features for processing [72]. This design enables CA to capture distance-dependent information along one spatial axis while preserving precise positional coordinates on the other. By integrating CA into our improved model, we effectively enhance the spatial and channel information of the input feature maps, leading to further performance improvements.

In our experiments, we integrated the CA attention module into the backbone network of the improved model, which resulted in enhanced feature representations on the original feature maps. The combination of the improved loss function and the attention mechanism allows the model to perform feature learning and weight adjustment more efficiently, particularly when addressing shape changes and occlusion issues in detection tasks.

IV. 4. Experimental Results and Validation

A. Presetting of Basic Conditions for Model Training

After the model environment configuration was set up and the dataset was labeled, the model was trained. The whole model was built on the Pytorch platform [73] based on Python 3.9, and some algorithms and programs for experiments were also established. Video data streams were converted into image data streams through Opencv. The processor based on Intel ® Core™ i5-12400f CPU@4.00 GHz was adopted for data loading, and NVIDIA GeForce RTX 3060@12.0 GB GPU with a memory of 32GB was employed to train the image dataset of the model.

The data set includes 7,400 images, with a size of 640x640x3, including an incomplete data set of 4,062 original images, and 3,338 subsequently semi-automatically labeled images. An incomplete data set refers to the data set of some category data that are difficult to acquire and have not been adjusted and balanced. The training set and validation set in the initial data set were set at 8:2, and 3,338 semi-automatically labeled images were required to form a test set to test the generalization ability of the PPE model. In addition, all the newly labeled 7,400 images were used as a new data set to be input into the model for training. The brand-new test set was a series of construction photo streams and video streams shot in the actual scene, and some categories were adjusted and balanced. This data set includes all kinds of background images of the construction site and a complete data set of images shot with different working postures of workers and under different conditions.

For parameter selection in the model training process, the pre-training mode was adopted in this paper. Based on the training results of YOLOv5 on the COCO dataset, different model configuration files and the corresponding pre-training weights were employed for training according to different models.

B. Training Based on an Incomplete Dataset

1) MODEL PARAMETER CONFIGURATION

In this paper, our objective is to test the effect of two models under two loss functions, resulting in a total of four model classes, based on the labeled benchmark dataset. Our analysis focuses on two models in the 6th branch of the latest YOLOv5 series: the largest model (Yolov5x) and the smallest model (Yolov5n). We aim to investigate the effect of incomplete datasets on the detection results of the PPE model, and the parameters of the models are shown in Table 2. For comparative experiments, we change the loss function and the model size to examine their actual impact on PPE detection in the case of incomplete datasets.

Furthermore, we incorporate an attention mechanism into the model Yn6S after improving the loss function. We compare models with different attention mechanisms to verify the effectiveness of the CA attention. The models with different attention mechanisms are named CA-Yn6SE, SE-Yn6SE, ECA-Yn6S, and CBAM-Yn6S. Other parameters are kept unchanged for the comparison.

By conducting these experiments and model comparisons, we aim to gain valuable insights into the impact of loss functions and model sizes on PPE detection and verify the effectiveness of attention mechanisms in enhancing the model's performance.

TABLE 2
MODELS BASED ON DIFFERENT LOSS FUNCTIONS AND DIFFERENT SIZES

NO.	MODEL NAME (ABBREVIATION)	EPOCHS	IMAGE SIZE	DEPTH_MULTIPLE	WIDTH_MULTIPLE
1	YOLOV5N_6.2-CIOU(YN6C)	50	640X640	0.33	0.25
2	YOLOV5X_6.2-CIOU(YX6C)	50	640X640	1.33	1.25
3	YOLOV5N_6.2-EIOU(YN6E)	50	640X640	0.33	0.25
4	YOLOV5X_6.2-EIOU(YX6E)	50	640X640	1.33	1.25
5	YOLOV5N_6.2-SIOU(YN6S)	50	640X640	0.33	0.25
6	YOLOV5X_6.2-SIOU(YX6S)	50	640X640	1.33	1.25
7	YOLOV5N_6.2-SEIOU(YN6SE)	50	640X640	0.33	0.25
8	YOLOV5X_6.2-SEIOU(YX6SE)	50	640X640	1.33	1.25

2) COMPARISON OF THE RESULTS OF THE IMPROVEMENT PART

(1) Comparison of Improved Loss Functions

To assess the impact of the proposed loss function improvements on detection accuracy, we evaluated the performance of eight models trained for four categories: Hat (H), Person (P), Reflective-clothes (R), and Other-clothes (O). The indicator parameters AP and mAP are shown in Table 3.

TABLE 3 INDICATORS OF THE TRAINING RESULTS OF THE 8 MODELS

	Yn6C	Yn6E	Yn6S	Yn6SE	Yx6C	Yx6E	Yx6S	Yx6SE
H(AP)	0.92	0.922	0.92	0.926	0.933	0.937	0.936	0.935

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P(AP)	0.769	0.769	0.775	0.774	0.814	0.81	0.794	0.816
R(AP)	0.898	0.898	0.894	0.901	0.911	0.915	0.921	0.921
O(AP)	0.872	0.871	0.874	0.88	0.879	0.884	0.888	0.881

The results indicate a noticeable improvement in detection accuracy after applying the improved loss function, with a 0.4% and 0.8% increase in the Hat and Reflective-clothes categories, respectively, and a 0.4% improvement in mAP. Notably, the improved SEIoU-based model outperforms the CioU-, EioU-, and SIoU-based models in detecting PPE, demonstrating the effectiveness of the improvement. The results indicate a noticeable improvement in detection accuracy after applying the improved loss function, with a 0.4% and 0.8% increase in the Hat and Reflective-clothes categories, respectively, and a 0.4% improvement in mAP. Notably, the improved SEIoU-based model outperforms the CioU-, EioU-, and SIoU-based models in detecting PPE, demonstrating the effectiveness of the improvement. This improvement is particularly impactful for detecting reflective clothing and other clothing categories, characterized by variable features and inconsistently labeled boxes in the image， and the position and aspect ratio of the rectangle box will change significantly. The SEIoU's penalty term effectively addresses these problems, leading to enhanced detection accuracy.

(2) Comparison of Adding Attention Mechanism

Table 4 below presents a comparison of the training results of the four models with the added attention mechanism.

TABLE 4
COMPARISON OF TRAINING RESULTS WITH THE ATTENTION MECHANISM

	SE	CBAM	ECA	CA
P	0.849	0.848	0.846	0.841
R	0.798	0.803	0.797	0.811
maAP	0.856	0.857	0.858	0.86

The YOLOv5 model with the added CA attention mechanism demonstrates higher recall compared to the other models, as shown in Figure 6. This improvement indicates that the model effectively captures positive samples, reducing missed detections. Given the nature of the PPE detection task, achieving a high recall is crucial to accurately identify all workers wearing PPE and minimize safety violations, such as the failure to wear reflective clothing.

Despite a slight decrease in precision, the YOLOv5 model with the addition of CA achieves the highest mAP among the different attentional mechanisms. This outcome suggests that the CA mechanism excels in striking a balance between accuracy and recall, making it the most suitable choice for PPE detection.

In summary, our evaluation demonstrates that the improved SEIoU loss function and the selected CA attention mechanism are more effective in YOLOv5 compared to other loss functions and attention mechanisms. As a result, we constructed the improved YOLOv5 model, integrating the improved SEIoU loss function and the CA attention mechanism. Subsequent tests on incomplete datasets confirmed the positive impact of these enhancements.

3) TRAINING EFFECT OF THE INCOMPLETE DATASET

The training results under the incomplete dataset can be analyzed in two ways

(1) Analysis of Model Indicator Parameters

In this section, we analyze the performance indicators of our detection model based on relevant index data. The precision, recall, and mAP of the Person, Reflective-clothes, and Other-clothes categories were examined among the four categories. Two main reasons contribute to the observed low precision in the Person and Reflective-clothes categories:

a. Labeling Imbalance:

To gain insights into the characteristics of our training dataset, we provide a detailed analysis depicted in Fig. 7. This figure includes:

1 Distribution of Training Dataset: The first image in Fig.5 illustrates the distribution of the training dataset, specifying the quantity of data for each category.

2 Bounding Box Dimensions: The second image in Fig.5 breaks down the sizes and quantities of the bounding boxes used in the training dataset.

3 Relative Positions of Object Centers: The third image visualizes the distribution of object centers relative to the entire image, offering insights into the spatial arrangement.

4 Aspect Ratios of Objects: The fourth image in Fig.5 depicts the ratios of height to width for objects within the images, providing information about the diversity of object shapes.

The Person category has approximately 2,000 labels, while the Reflective-clothes category has around 2,800 labels. Both of these categories have lower labeling counts compared to the other two categories.

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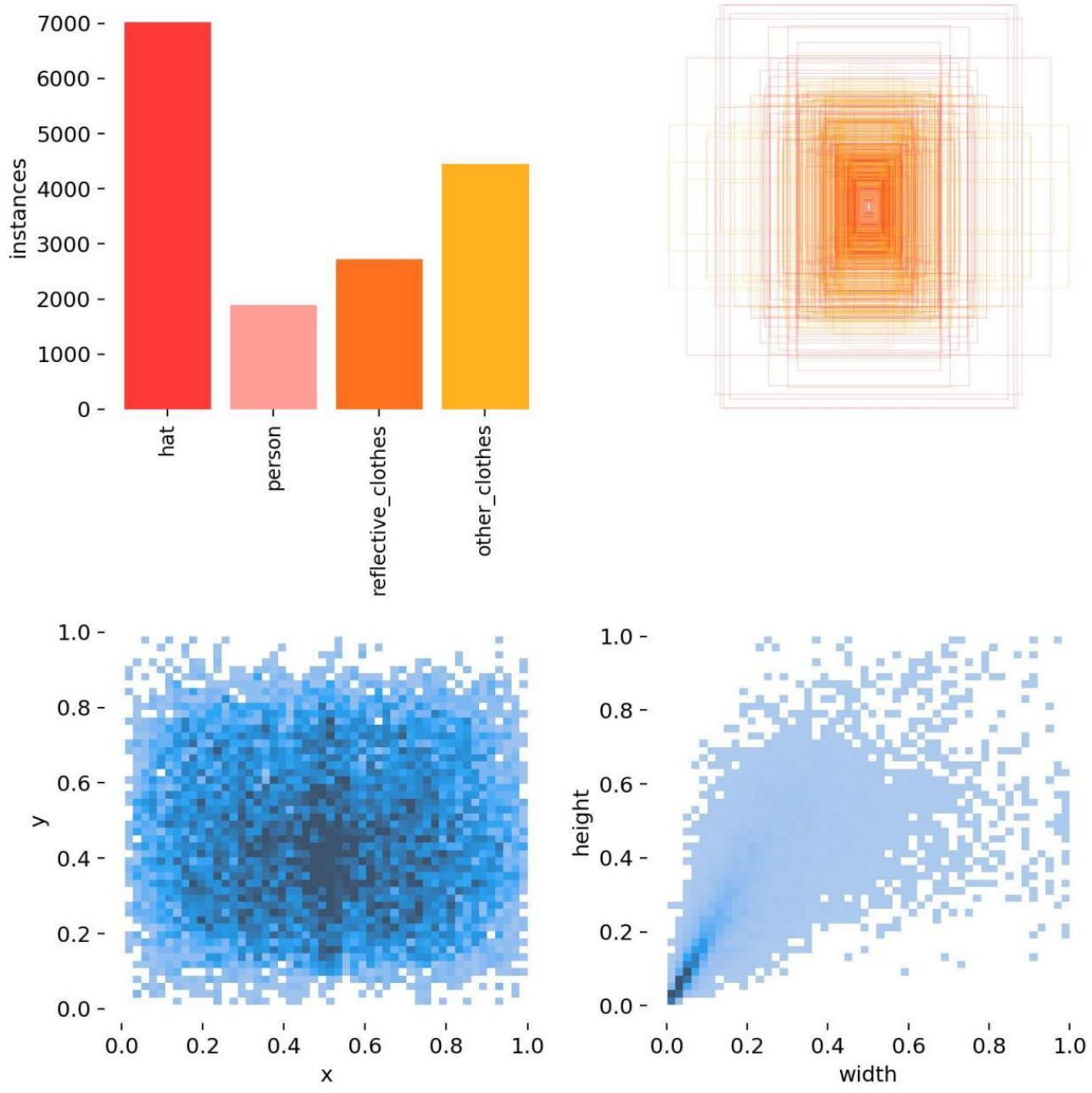


FIGURE 5. **Statistical Graph of Labels of Each Category**

During data collection, our focus was on acquiring images from actual construction sites or similar scenes where workers predominantly wear hard hats. As a result, the number of labels for the Person category in images with workers not wearing hard hats is significantly smaller, leading to a lower object detection precision.

b. Feature Overlapping: Detection of hard hats benefits from distinct image features, resulting in high precision and recall rates. However, distinguishing between workers wearing reflective clothing and other types of clothes poses challenges due to overlapping features. The limited number of labeled samples for workers wearing reflective clothing, coupled with the overlapping features, makes precise labeling difficult.

Addressing these challenges is crucial to improving the overall precision of object detection in the specified categories.

(2) Analysis of Actual Detection Effect

In terms of the actual detection effect based on an incomplete dataset, certain problems are evident in the Person category and the Reflective-clothes category, particularly missed detections and false detections, while there are fewer problems of false detection in the Hat category, as illustrated in Fig.6.

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FIGURE 6. **Detection Effect Diagram of the Training Based on an Incomplete Dataset**

Two primary reasons account for these issues:

I. The challenges in the Person category arise from its small size and relatively indistinct features in images compared to hard hats. In this context, "small size" refers to the relatively diminutive representation of hard hats and individuals not wearing reflective clothing in construction site images, as opposed to reflective clothing. These instances often pose difficulties in detection due to their smaller pixel areas and reduced spatial extent within the image frame.

The features of the "small size" category, particularly those related to workers wearing hard hats and not wearing reflective clothing, tend to blend with environmental elements, further complicating the detection process. During the labeling process, interference from the Other-clothes category may occur. For instance, when labeling the Other-clothes category, the rectangular bounding box may inadvertently include instances from the Person category, leading to interference between these two categories during subsequent detection processes.

Moreover, in conditions of poor image clarity and insufficient lighting, workers wearing specific clothes, such as black attire, may resemble workers facing away from the camera. In these scenarios, the smaller features of the Person category might be mistakenly identified as belonging to the Other-clothes category. Due to the dataset's imbalance, the number of samples exhibiting these interference conditions is limited, making accurate detection challenging.

However, considering that both not wearing hard hats and not wearing reflective clothing are classified as unsafe worker behaviors, the interference of the Person category by the Other-clothes category results in the misidentification of one unsafe state as another. This misidentification is deemed more acceptable compared to erroneously identifying instances of not wearing reflective clothing as wearing reflective clothing, which is a critical safety concern.

II. Detecting the Reflective-clothes category and the Other-clothes category is also challenging. Some features may be obscured, and improper wearing of reflective clothing may cause the computer to detect features of both reflective clothing and the original clothes simultaneously, making the detection of reflective clothing difficult. Reflective clothing is usually worn as clothes over the workers' regular attire and does not cover the original clothes. This can lead to the computer mistakenly identifying the information of reflective clothes as that of the underlying clothes during identification. Moreover, workers exhibit various construction postures and movements on construction sites, such as squatting, sitting, and moving, which can cause deformations in reflective clothing. These postures and movements change the information captured in images, making it challenging for the computer to accurately recognize the Reflective-clothes category and the Other-clothes category.

Due to the imbalance of the dataset and the incomplete labeled information acquired from construction sites, the distinct feature information of each category becomes less evident. Expanding the basic dataset to include images under various conditions is necessary to address these issues.

C. Training Based on a Complete Dataset

1) CONSTRUCTION OF A COMPLETE DATASET BY SEMI-AUTOMATIC LABELING

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On the basis of the incomplete dataset, a complete dataset was constructed using the semi-automatic labeling approach proposed in Section 3.1. The semi-automatic labeling technology helps in continuously forming labeled pictures based on object detection in construction site images. To enhance the accuracy and efficiency of the process, the CA-Yx6SE model, known for its higher detection accuracy and speed compared to the Yn6SE model, was utilized for auto-labeling.

After auto-labeling, the labeling results were carefully checked and adjusted to eliminate cases of missed detection, false detection, and repetition of detection boxes. The new datasets contained 3,338 additional images, combined with the original 4,062 images, resulting in a total of 7,400 images. These images were then divided into a training set of 5,819 images and a validation set of 1,581 images, providing comprehensive coverage of various construction situations and conditions on the construction site.

The adjustments made during the labeling process focused on resolving issues arising from mutual interference between the Reflective-clothes category and the Other-clothes category, as well as false detections caused by interference with the Person category by the Other-clothes category.

By following this approach, we successfully constructed a complete dataset that will be crucial for further analysis and improvements in PPE detection accuracy in construction environments.

2) TRAINING EFFECT OF THE BRAND-NEW DATASET

The new dataset was input into the model to complete a new round of training, and the models trained based on the complete dataset were named CA-Yn6SEC and CA-Yx6SEC. The basic parameter configuration was the same as that in Section 4.2, as shown in Table 2, and the setting of specific parameters was the same as mentioned above. The model parameters and mAP under different datasets were compared, as shown in Table 5. The precision, recall, and mAP in model training results of the 4 categories are shown in Table 6.

TABLE 5
PARAMETERS AND PERFORMANCES OF THE MODELS UNDER DIFFERENT DATASETS

Number	Model categories	Dataset size (sheets)	Image size	mAP (0.5)	Speed-GPU (FPS)	Speed-detect(ms)
1	CA-Yn6SE	4062	640x640	0.870	172.4	5
2	CA-Yx6SE	4062	640x640	0.894	38.5	5
3	CA-Yn6SEC	7400	640x640	0.916	172.4	25
4	CA-Yx6SEC	7400	640x640	0.938	38.5	25

TABLE 6
MODEL TRAINING EFFECT BASED ON PPE COMPLETE DATASET

	H			P			R			O		
	P	R	AP	P	R	AP	P	R	AP	P	R	AP
Yn6SE	0.898	0.884	0.93	0.84	0.76	0.82	0.876	0.856	0.91	0.841	0.766	0.82
Yn6SEA	0.923	0.886	0.938	0.901	0.8	0.885	0.911	0.868	0.938	0.875	0.834	0.901
Yx6SE	0.934	0.899	0.952	0.848	0.838	0.871	0.918	0.862	0.904	0.857	0.79	0.848
Yx6SEA	0.94	0.909	0.952	0.948	0.898	0.946	0.934	0.911	0.943	0.888	0.873	0.909

(1) Analysis of Model Index Parameters

It can be seen that, regardless of the model size, training time, and other reasons, the average mAP of CA-Yx6SEC with a network depth of 1.33 and network width of 1.25 is the highest, and AP values of the four categories are 0.9542, 0.946, 0.943 and 0.909 respectively. All relevant indexes of the model are superior to those of other models. The training effect based on the brand-new dataset has been significantly improved, and the precision and recall in the Person(P) category and the Other-clothes(O) category have been improved by more than 6%.

Specifically, the average precision, average recall, and mAP@0.5 of CA-Yx6SEC are improved by 3.8%, 5.1%, and 4.4% respectively compared with the above indexes of Yx6S. Besides, the average precision, average recall, and mAP@0.5 of CA-Yn6SEC are improved by 3.9%, 3.1%, and 4.6% respectively compared with these indexes of Yn6SE. Especially in the Reflective-clothes category and the Person category with a low detection precision before, mAP@0.5 is improved to a great extent. The newly added datasets have greatly improved the detection effect of workers not wearing hard hats and reflective clothing. The results indicate that the complete dataset can facilitate achieving a better PPE detection effect than the unbalanced dataset.

In addition, the CA-Yn6SE model with the lightest weight and the smallest model parameters achieves the mAP of 91.6%, which also reveals that the YOLOv5_6.2-CA-SEIoU model based on the brand-new dataset has outstanding performance and can achieve excellent PPE detection effect. If CA-Yn6SE is adopted, it can obviously improve the effectiveness in detecting the categories of small objects such as the Hat category and the Person category, and the detection precision in the Hat category is as high as 95.2%. Therefore, in view of the precision, it is advisable to choose the Yx6SE model during the comprehensive PPE detection of various categories. If speed is highlighted, or only the categories of reflective clothing are considered, Yn6SE can undoubtedly achieve an extremely high speed and precision.

The test set based on video streams further validates the effectiveness of the brand-new dataset for real-time detection on construction sites. In terms of detection precision, the four categories can basically be correctly labeled, and the correctness detection of Reflective-clothes and its opposite category is also greatly improved. Although the improvement

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of the Other-clothes category is obvious, the accuracy of the Other-clothes category is not high enough compared with other categories. A significant reason is that the situation of wearing other clothes is complicated, and the feature shapes of various clothes in the image are variable in different scenes. Since the case of wearing other clothes is more complicated and the feature shapes of various clothes in the image are variable in different scenes, its correct detection effect is more difficult.

As to detection speed, in RTX 3060 GPU environment, the detection speed of the CA-Yn6SE model is up to 172.4 FPS, which is greatly improved compared with the 38.5 FPS of CA-Yx6SE, and the mAP is only reduced by 2%. The results show that the CA-Yn6SEC model based on the brand-new dataset can meet the real-time and high-precision requirements of PPE detection on construction sites. However, in case of harsh requirements for implementation, the CA-Yx6SEC model can also achieve the effectiveness of PPE detection with higher precision.

(2) Analysis of Actual Detection Effect

For analysis of the actual effect of the improved CA-SEIoU, the model with the highest detection speed was selected. Yn6C and CA-Yn6SE trained based on the brand-new dataset were compared, and reasoning detection was performed based on this model. The relevant data are taken from 5 video streams shot on the spot, and part of the detection results are shown in Fig.7.

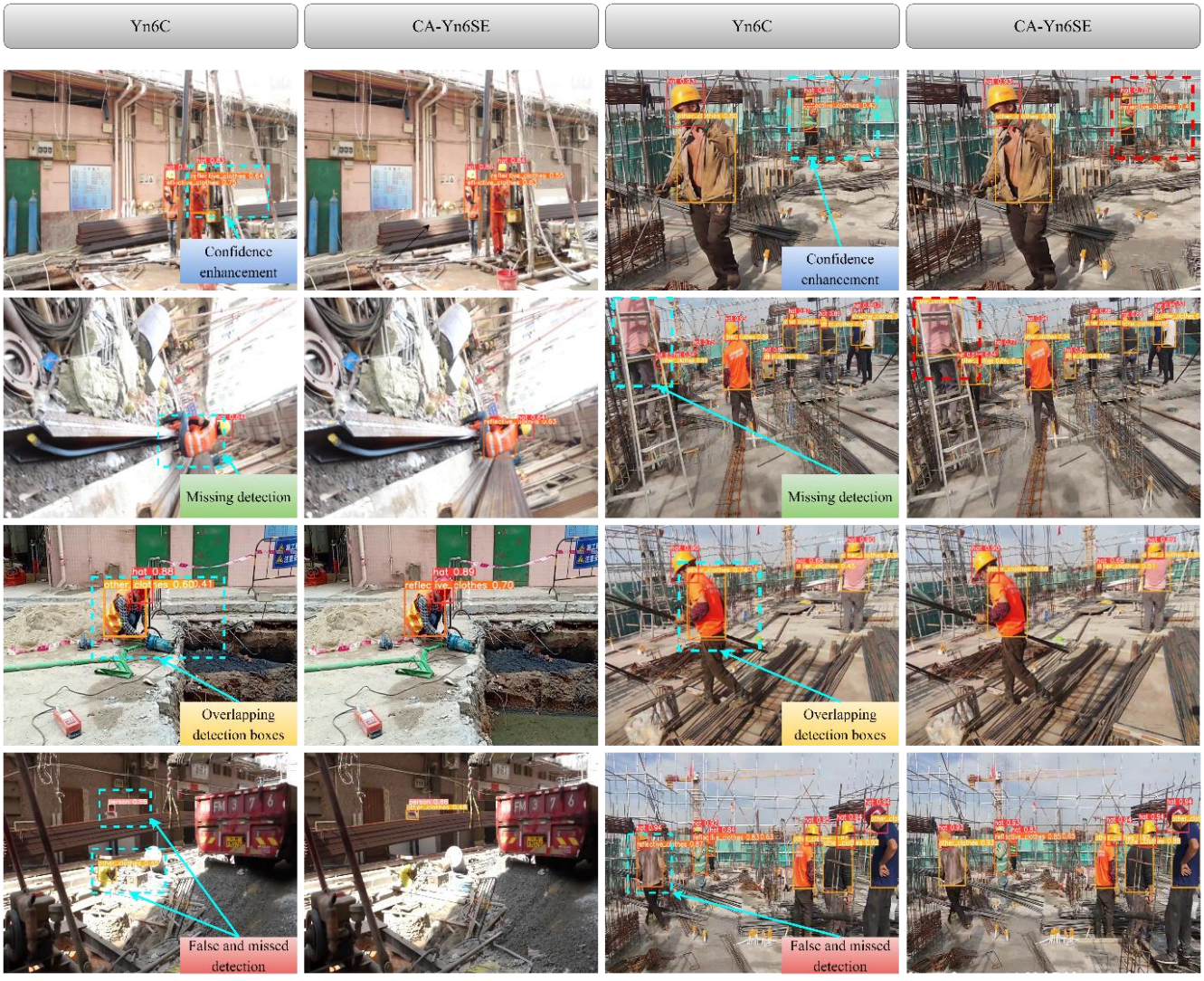


FIGURE 7. Reasoning Effect of Model Detection Based on CIoU/CA-SEIoU

For analysis of the actual effect of the improved CA-SEIoU, we selected the model with the highest detection speed, which was named CA-Yn6SE. We then compared CA-Yn6SE with the baseline model Yn6C, both trained based on the brand-new dataset. Reasoning detection was performed based on this model using relevant data from 5 video streams, and part of the detection results is shown in Fig.10.

a. Confidence enhancement: In the same detection results (First row in the figure), the detection box of the model based on CA-SEIoU has higher confidence.

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b. Missed detection: When the angle of the image changes or the features are blocked (Second row), the CA-SEIoU-based model can detect these features, while the CIoU-based model misses the Reflective-clothes category and the Other-clothes category respectively.

c. Overlapping of detection boxes: When the CIoU-based model detects the Reflective-clothes category and the Other-clothes category, two kinds of category boxes (Third row) are labeled on a feature at the same time. The reason is that the features of the Other-clothes category interfere with those of the Reflective-clothes category, and the two kinds of features appear in the detection box at the same time. In the practical labeling, it is considered that some features of the Other-clothes category appear in the labeling box of the category of workers wearing reflective clothing normally, and they are also labeled as the R category. In this regard, the model based on CA-SEIoU achieves a better detection effect, which is basically the same as the expected result, This also effectively solves the overlap problem prevalent in detecting PPE.

d. False detection: When the features are obviously blocked, the CA-SEIoU-based model can still correctly detect the Reflective-clothes category (Fourth row). The CIoU-based method mistakenly detects environmental features on the spot as the Other-clothes category with a confidence of 0.45, while the CA-SEIoU-based method detects the four categories normally.

The CA-SEIoU-based model (CA-Yn6SE) significantly improves the detection effects on reflective clothing and its opposite category. It excels in PPE detection under various complex backgrounds and exhibits certain generalization abilities. This indicates that the balanced and expanded dataset with high-quality labeling contributes to achieving a high mAP in PPE detection.

However, the CA-SEIoU-based model still faces challenges in detecting the category of workers not wearing reflective clothing, and its detection effect is slightly lower than that for hard hats. Many false detections occur in the detection of the Reflective-clothes category and the Other-clothes category. This is attributed to the complexity of the not-wearing-reflective- clothing category, where the features of other clothes significantly outnumber those of the other three categories in the dataset. Moreover, the model struggles to distinguish between ordinary clothes and reflective clothing when features from both categories appear in the labeling box during detection.

Overall, the CA-SEIoU-based model shows promise for improving PPE detection, particularly for reflective clothing. However, further enhancements and refinements are required to achieve higher precision in detecting workers not wearing reflective clothing, and overcoming the challenges posed by feature overlap in the dataset.

3) ABLATION EXPERIMENTS

To assess the impact of the attention mechanism and the improved loss function proposed in this paper on network performance, we conducted ablation experiments using the complete dataset. The experiments involved three model configurations: the original YOLOv5 model (Baseline), the YOLOv5 model with the added attention mechanism (A-YOLOv5), and the YOLOv5 model with the improved loss function (L-YOLOv5). Additionally, we combined both the attention mechanism and improved loss function in the YOLOv5 model (AL-YOLOv5). The experimental results and detection results are shown in Table 7.

TABLE 7
ABLATION EXPERIMENT RESULTS ON COMPLETE DATASET

	Baseline	A-YOLOv5	L-YOLOv5	AL-YOLOv5
P	0.846	0.855	0.866	0.855
R	0.797	0.806	0.800	0.811
mAP	0.857	0.858	0.860	0.860

(1) A-YOLOv5 model: Comparing A-YOLOv5 with the baseline model, we observed no significant change in overall mAP. However, as shown in Fig.8, the A-YOLOv5 model demonstrated improved detection of overlapping frames and more accurate localization of PPE equipment, especially in complex scenarios involving workers wearing reflective clothing and other garments simultaneously. The attention mechanism significantly enhanced PPE detection performance.

(2) Comparison with AL-YOLOv5 and L-YOLOv5:

Acknowledging that quantitative results might not fully capture qualitative improvements, This will provide a more detailed breakdown of specific scenarios where the attention mechanism and improved loss function contribute to enhanced detection accuracy, reduced frame overlap, and minimized missed and false detections.

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FIGURE 8. **Comparison of Detection Results of A-YOLOv5 and Baseline Models**

1) L-YOLOv5 model: Comparing L-YOLOv5 with the baseline model, the mAP metric improved by 0.4 percentage points. The improved loss function introduced a shape penalty term for reflective clothing features, resulting in enhanced detection accuracy for reflective clothing. However, the improvement was limited in detecting frame overlap.

2) AL-YOLOv5 model: Comparing AL-YOLOv5 with A-YOLOv5, we observed an mAP of 86%, comparable to A-YOLOv5. However, as shown in Fig.9, the detection performance of AL-YOLOv5 surpassed L-YOLOv5, showing significant improvements in detection accuracy, reduction of frame overlap issues, missed detections, and false detections.

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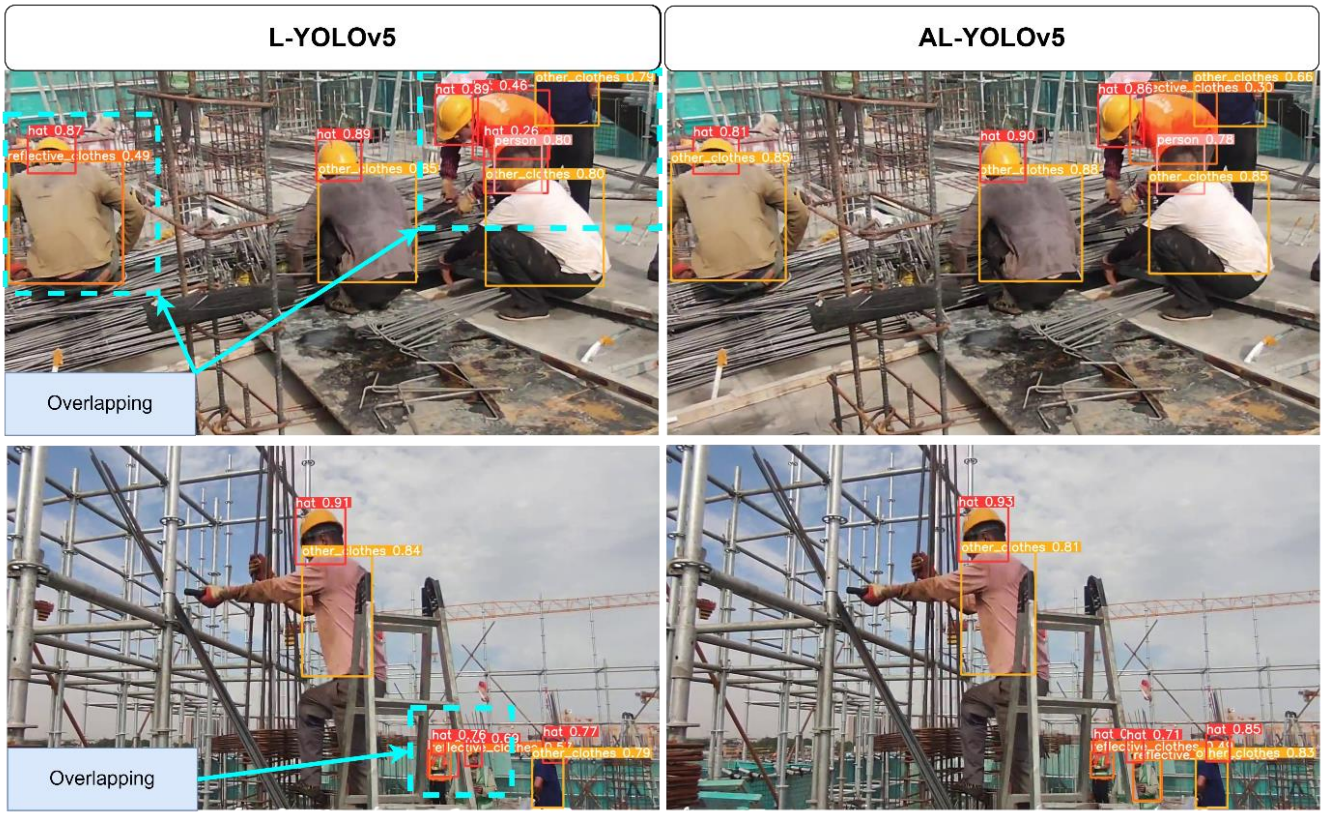


FIGURE 9. Comparison of Detection Results of AL-YOLOv5 and L-YOLOv5 Models

In summary, the integrated model (AL-YOLOv5) with both the attention mechanism and improved loss function outperformed the other configurations, offering improved detection accuracy and effectively solving the challenges related to reflective clothing detection. Emphasizing the practical implications of our method is crucial. Beyond numerical metrics, the reduction in missed and false detections significantly enhances the reliability of our model in real-world applications. Through ablation experiments, we validated the performance improvements of the attention mechanism and improved loss function on the YOLOv5 model, highlighting their synergistic effects in PPE detection. These enhancements are vital for improving the accuracy and practicality of PPE detection, providing a more effective safety management tool for construction sites and similar environments.

V. DISCUSSION

A. Outstanding contributions to research findings

In this study, we discuss current trends and limitations in PPE detection and propose novel solutions to improve the accuracy and robustness of construction site safety hard hats and reflective clothing detection. To facilitate automated detection management, we constructed a novel dataset containing a variety of angles, backgrounds, and scenes for the four PPE classes. Using semi-automated annotation techniques, we achieve fast and accurate annotations that significantly reduce image annotation time and facilitate automation of detection management. The models trained on this dataset show sufficient generalization for automated safety management studies on construction sites, opening up new avenues for future research. By combining a solution for creating comprehensive datasets with an attention mechanism and an improved loss function, our research makes a significant contribution to the field of PPE detection.

By training on incomplete and complete datasets, we compare the effectiveness of the improved and baseline models for PPE detection. The results show that the AL-YOLOv5 model improves about 0.9 AP in the data-limited category and about 0.4 mAP in the four categories. It is worth noting that the improved model outperforms the baseline model, especially for challenging categories such as "Reflective-clothes" and "Other-clothes," where features are deformable and may cause interference. The attention mechanism significantly improves the detection of reflective clothing, reducing false detections, overlapping bounding boxes, and missed detections.

Compared with the original YOLOv5, our proposed model based on the improved loss function and attention mechanism provides great practical value in PPE detection. This approach not only addresses the shortcomings of existing methods but also enhances the overall performance of the YOLOv5 model, paving the way for a more efficient and reliable PPE detection system. Furthermore, the impact of our work is not limited to construction safety, as it can be applied to a variety of real-world scenarios where PPE compliance needs to be accurately and effectively monitored. Overall, our findings have great

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potential to enhance safety practices and prevent accidents in a variety of industrial and occupational environments.

B. Comparison with existing research results

By comparing our study to existing research, we aim to shed light on the limitations of previous studies. Existing research primarily addresses issues related to dataset limitations and suboptimal practical detection results. In particular, Zheng and Nath et al. achieved only 70% accuracy in detecting workers not wearing PPE. In addition, Fang [74] pointed out that workers' clothing can cause interference in detecting safety belts in practical scenarios, and Fang [75] thought that the colors and shapes of workers' clothing can significantly reduce the effectiveness of object detection.

In contrast, our study significantly mitigates these challenges through strategic approaches, including dataset expansion, loss function optimization, and the incorporation of an attention mechanism. Specifically, our study achieves an average mAP above 90% across four categories, with the lowest category achieving a detection rate of 80%. This significant improvement effectively narrows the detection gap between categories.

Our study makes a significant contribution to overcoming the challenge of sample imbalance, particularly in the detection of reflective clothing. Unlike previous studies that considered hard hats and reflective clothing as similar feature categories, our research addresses the unique nature of reflective clothing. This category is prone to variation with worker posture, and its features may interfere with the analysis of opposing categories in the dataset. As a result, our findings hold promise for advancing research on object category detection in real-world construction sites.

In addition, previous research falls short of providing an in-depth analysis of real-world scenarios. For example, Fang [74] noted the limitations of using only 693 images for model training, which hindered seat belt detection in certain contexts. Similarly, the efficacy of PPE detection studies based on laboratory video has yet to be validated in real-world scenarios [36]. In contrast, our study considers real-world engineering scenarios from the outset of dataset construction. This approach includes consideration of the impact of multiple worker poses on the detection of reflective clothing. While Fang [37] et al. collected over 100,000 images for worker and hard hats detection, they overlooked the unique challenges posed by reflective clothing and other variations in different scenarios. This is where reflective clothing differs significantly from hard hats detection. In addition, Chen [76] addressed the issue of object occlusion on construction sites by improving models based on the MOCS dataset. However, he also emphasized the need for further dataset enrichment to provide higher-quality training samples. Our study effectively fills this research gap.

Compared to research conducted under ideal laboratory conditions, our study prioritizes effective detection in challenging real-world engineering environments. We consider scenarios involving workers wearing hard hats and reflective clothing in various conditions, constructing a new dataset to enhance the detection of reflective clothing and hard hats at construction sites. This not only augments the usability of datasets and computational methods in real engineering but also aligns with the real-time nature of on-site detection.

This paper focuses on solving the challenge of PPE detection in real-world engineering contexts, highlighting the uniqueness of different PPE in engineering scenarios. Real-world engineering introduces new complexities to numerous PPE devices, including issues related to feature deformation and occlusion. Achieving satisfactory detection and recognition results becomes a challenging task, especially when detecting worker behavior. Overcoming these inherent challenges requires a comprehensive approach that encompasses both datasets and models, with dataset expansion and model optimization essential.

By addressing the complex challenge of recognizing PPE, including reflective clothing, in real-world scenarios, our research not only provides the engineering community with powerful tools but also opens up new prospects for cross-domain applications. For example, our newly developed dataset and refined methods hold great promise in the fields of surveillance and safety management. By focusing on real-world scenarios, we significantly improve the accuracy of reflective clothing detection. This not only enhances worker safety but also provides surveillance systems with the ability to efficiently detect PPE - ultimately raising safety standards and minimizing the risk of accidents. At the same time, our method serves as a valuable safety management tool, facilitating real-time monitoring of PPE compliance and improving overall management practices. Beyond surveillance and security, our research significantly contributes to the formalization of engineering knowledge and the advancement of computational methods. This work not only aids researchers in addressing challenges in PPE detection but also extends its impact to similar real-world problems in various engineering domains.

For instance, consider the field of defect detection, where image irregularities and smaller features can lead to misclassification [77]. The same attentional mechanisms employed in our reflective clothing detection can significantly enhance the handling of such issues. Likewise, several studies have highlighted the challenges posed by the coexistence of multiple target categories with varying aspect ratios and other morphological features like texture, size, and color. The addition of attentional mechanisms has proven effective in addressing these difficulties [78]. This mirrors our case with reflective clothing and non-wearing hard hats in construction projects, where the overlap of features and interference are addressed by refining the loss function and integrating attentional mechanisms, resulting in significantly improved detection accuracy.

In conclusion, our research extends beyond the confines of the engineering field, offering novel computational methods for addressing real-world challenges. These solutions hold

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vast potential, reaching into multiple fields and opening doors to a diverse array of applications.

C. Limitations of the study and prospects for future research

Despite their dynamic deformation properties, reflective clothing is often overlooked in deep learning-based approaches to worker PPE detection, with the focus primarily on hard hats. However, the detection of reflective clothing poses significant challenges due to their susceptibility to variation and interference from ordinary clothes. We argue that reflective clothing should be as important as hard hats and PPE, and should be automatically detected using deep learning models. In addition, analysis of data classes that are confounded by comparison objects, such as reflective clothing, can provide valuable insights for similar object recognition studies. To address the challenges posed by the complexity and diversity of objects on construction sites, it is critical to collect comprehensive datasets and accurate annotations.

Although this paper proposes an attention mechanism and an improved loss function to improve the performance of PPE detection, there are still some limitations to consider. First, although ablation experiments were conducted on the complete dataset to verify the effectiveness of the attention mechanism and the improved loss function, more experiments are still needed to verify the generalization performance of these methods in different scenarios and conditions. Secondly, due to the complex and variable environment of construction sites, factors such as workers' postures and occlusions can affect the detection of reflective clothing, so the detection effect for some special cases still needs to be further improved. In addition, the dataset used in this paper may still have certain imbalance problems, resulting in poor model performance in some categories, such as when detecting workers not wearing hard hats, due to the inconspicuous features, it is easy to merge with the environmental background and other categories, interfere with other problems, resulting in the detection accuracy is still to be improved, and it is necessary to further optimize the data collection, the sample balancing strategy, and further model improvement to solve this problem.

Our study confirmed the YOLOv5 model's high precision and speed in detecting both reflective clothes and safety hard hats. To extend the scope of our findings, we recognize the significance of exploring the detection performance of the latest YOLO version. In our future research, we plan to conduct comparative analyses that highlight the applications of the newest models in construction scenarios. This comparative analysis will consider factors such as computational efficiency and detection accuracy, providing a comprehensive view of the evolving landscape of object detection in construction safety. Future research directions can focus on optimizing the design of the attention mechanism and exploring more effective ways to emphasize the key features of reflective clothing and hard hats to further improve detection accuracy. In addition, introducing more data enhancement techniques can increase dataset diversity and improve model generalization. Considering the comprehensive analysis of OD from multiple perspectives can enrich the validity of PPE detection and amplify its practical impact. Furthermore, considering the association between head (hard hats) and body (reflective clothing) detection in PPE detection may lead to more practical and error-free detection. In conclusion, our proposed method has the potential to provide more effective safety management solutions for construction sites and find wider applications in various fields. Through further in-depth research and optimization, we aspire to make a significant contribution to safety practices and accident prevention in industrial and occupational environments.

VI. CONCLUSION

In this study, we propose an integrated approach to improve PPE detection at construction sites, which is achieved by creating diverse PPE datasets, introducing an attention mechanism, and improving the SEIoU loss function. Our experimental results demonstrate the practical value of this method in ensuring worker safety and efficient PPE management. Specific conclusions are as follows:

(1) Improved model performance: By replacing the SEIoU loss with the CIoU loss in YOLOv5 and incorporating the attention mechanism, our improved model exhibits higher accuracy in several categories, resulting in a 0.4% improvement in mAP compared to the original model. In particular, the AP improvement for categories with limited data labeling, such as not wearing a hard hat and wearing reflective clothing, reaches approximately 0.9%, outperforming the original model even in challenging detection scenarios. The improvement of the attention mechanism and the loss function has led to a significant improvement in the four effects of the model, such as confidence, missed detection, misdetection, and overlapping detection frames.

(2) Dataset expansion and generalization: Using a semi-automated approach combining manual labeling and automatic model labeling, we successfully extended the PPE detection dataset, providing sufficient data support for training the improved model on a brand-new dataset. The improved model shows robust generalization performance on diverse construction site images, achieving over 90% mAP and the highest detection speed of 172.4, ensuring high accuracy and meeting real-time requirements.

(3) Challenges and future research: Our results highlight the challenges associated with the detection of reflective clothing, an essential PPE device such as hard hats. The deformable and easily obscured features of reflective clothing, along with potential overlap with features of other clothing categories, lead to detection errors such as missed detection, false detection, and overlapping detection frames. Future research efforts should focus on overcoming these challenges in detecting this category with dynamic features and require further investigation.

In conclusion, our research makes a significant contribution to the field of PPE detection in construction sites. Our integrated approach not only improves detection accuracy and real-time

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performance but also points out new research directions for PPE detection. We believe that these findings will lead to more effective safety management practices and offer versatile applications in various fields.

REFERENCES

[1] B. H. W. Guo, Y. Zou, Y. Fang, Y. M. Goh, and P. X. W. Zou, "Computer vision technologies for Saf. Sci. and management in construction: A critical review and future research directions", *Saf. Sci.*, vol. 135, p. 105130, Mar. 2021.

[2] E. A. L. Teo, F. Y. Y. Ling, and A. F. W. Chong, "Framework for project managers to manage construction safety" *Int. J. Project Manage.*, vol. 23, no. 4, pp. 329-341, May 2005.

[3] J. W. Hinze and J. Teizer, "Visibility-related fatalities related to construction equipment", *Saf. Sci.*, vol. 49, no. 5, pp. 709-718, Jun. 2011.

[4] P. E. D. Love, P. Teo, J. Smith, F. Ackermann, and Y. Zhou, "The nature and severity of workplace injuries in construction: engendering operational benchmarking", *Ergonomics*, vol. 62, no. 10, pp. 1273-1288, Oct. 2019.

[5] I. W. H. Fung, Y. Y. Lee, V. W. Y. Tam, and H. W. Fung, "A feasibility study of introducing chin straps of safety helmets as a statutory requirement in Hong Kong construction industry", Saf. Sci., vol. 65, pp. 70-78, Jun. 2014.

[6] H. Wu and J. Zhao, "An intelligent vision-based approach for helmet identification for work safety" *Comput. Ind.*, vol. 100, pp. 267-277, Sep. 2018.

[7] K. Shrestha, P. P. Shrestha, D. Bajracharya, and E. A. Yfantis, "Hard-Hat Detection for Construction Safety Visualization", *J. Constr. Eng.*, vol. 2015, p. 721380, Feb. 2015.

[8] N. D. Nath, A. H. Behzadan, and S. G. Paal, "Deep learning for site safety: Real-time detection of personal protective equipment", *Autom. Constr.*, vol. 112, p. 103085, Apr. 2020.

[9] M. Zhang, R. Shi, and Z. Yang, "A critical review of vision-based occupational health and safety monitoring of construction site workers", *Saf. Sci.*, vol. 126, p. 104658, Jun. 2020.

[10] S. Tang and M. Golparvar-Fard, "Joint Reasoning of Visual and Text Data for Safety Hazard Recognition", *Comput. Civ. Eng. 2017*, pp. 450-457, Jun. 2017.

[11] S. Chi and C. H. Caldas, "Image-Based Safety Assessment: Automated Spatial Safety Risk Identification of Earthmoving and Surface Mining Activities", *J. Constr. Eng. Manage.*, vol. 138, no. 3, pp. 341-351, Mar. 2012.

[12] F. Al-Sahar, A. Przegalińska, and M. Krzemiński, "Risk assessment on the construction site with the use of wearable technologies", *Ain Shams Eng. J.*, vol. 12, no. 4, pp. 3411-3417, Dec. 2021.

[13] J. Jeon and H. Cai, "Classification of construction hazard-related perceptions using: Wearable electroencephalogram and virtual reality", *Autom. Constr.*, vol. 132, p. 103975, Dec. 2021.

[14] S. Paneru and I. Jeelani, "Computer vision applications in construction: Current state, opportunities & challenges", *Autom. Constr.*, vol. 132, p. 103940, Dec. 2021.

[15] K. Li, X. Zhao, J. Bian, and M. Tan, "Automatic Safety Helmet Wearing Detection", 2017 IEEE 7th Annual International Conference on CYBER Technology in Automation, Control, and Intelligent Systems (CYBER), pp. 617-622, Jul. 2017

[16] A. H. M. Rubaiyat et al., "Automatic Detection of Helmet Uses for Construction Safety", *IEEE/WIC/ACM International Conference on Web Intelligence Workshops (WIW)*, no. 13-16, pp. 135-142,), Oct. 2016.

[17] H. Zhang, X. Yan, H. Li, R. Jin, and H. Fu, "Real-Time Alarming, Monitoring, and Locating for Non-Hard-Hat Use in Construction", *J. Constr. Eng. Manage.*, vol. 145, no. 3, p. 04019006, Mar. 2019.

[18] M.-W. Park, N. Elsafty, and Z. Zhu, "Hardhat-Wearing Detection for Enhancing On-Site Safety of Construction Workers", *J. Constr. Eng. Manage.*, vol. 141, no. 9, p. 04015024, Sep. 2015.

[19] J. Wu, N. Cai, W. Chen, H. Wang, and G. Wang, "Automatic detection of hardhats worn by construction personnel: A deep learning approach and benchmark dataset", *Autom. Constr.*, vol. 106, p. 102894, Oct. 2019.

[20] B. E. Mneymneh, M. Abbas, and H. Khoury, "Vision-Based Framework for Intelligent Monitoring of Hardhat Wearing on Construction Sites", *J. Comput. Civ. Eng.*, vol. 33, no. 2, p. 04018066, 2019.

[21] D. Shan, M. Shehata, and W. Badawy, "Hard hat detection in video sequences based on face features, motion and color information", *2011 3rd International Conference on Computer Research and Development*, vol. 4, pp. 25-29, Mar. 2011.

[22] J. Shen, X. Xiong, Y. Li, W. He, P. Li, and X. Zheng, "Detecting safety helmet wearing on construction sites with bounding-box regression and deep transfer learning", *Comput.-Aided Civ. Infrastruct. Eng.*, vol. 36, no. 2, pp. 180-196, 2021.

[23] X. Chen, Y. Li, and Y. Nakatoh, "Pyramid attention object detection network with multi-scale feature fusion", *Comput. Electr. Eng.*, p. 108436, Dec. 2022.

[24] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-Excitation Networks", *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 8, pp. 2011-2023, Aug. 2020.

[25] S. Woo, J. Park, J.-Y. Lee and I. S. Kweon, "CBAM: Convolutional block attention module", *Proc. Eur. Conf. Comput. Vis. (ECCV)*, pp. 3-19, Sep. 2018.

[26] G. Iannizzotto, L. Lo Bello, and G. Patti, "Personal Protection Equipment detection system for embedded devices based on DNN and Fuzzy Logic", *Expert Syst. Appl.*, p. 115447, Dec. 2021.

[27] X. Ke, W. Chen, and W. Guo, "100+ FPS detector of personal protective equipment for worker safety: A deep learning approach for green edge computing", *Peer Peer Netw. Appl.*, vol. 15, no. 2, pp. 950-972, Mar. 2022.

[28] J. P. Cheng, P. K.-Y. Wong, H. Luo, M. Wang, and P. H. Leung, "Vision-based monitoring of site safety compliance based on worker re-identification and personal protective equipment classification", *Autom. Constr.*, p. 104312, Jul. 2022.

[29] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal Visual Object Classes (VOC) Challenge", *Int. J. Comput. Vision* , vol. 88, no. 2, pp. 303-338, Jun. 2010.

[30] T.-Y. Lin et al., "Microsoft COCO: Common Objects in Context", *Proc. Eur. Conf. Comput. Vis. (ECCV)*, pp. 740-755, May 2014

[31] A. Xuehui, Z. Li, L. Zuguang, W. Chengzhi, L. Pengfei, and L. Zhiwei, "Dataset and benchmark for detecting moving objects in construction sites", *Autom. Constr.*, no. 122, p. 103482, Feb. 2021.

[32] D. Roberts and M. Golparvar-Fard, "End-to-end vision-based detection, tracking and activity analysis of earthmoving equipment filmed at ground level", *Autom. Constr.*, no. 105, p. 102811, Sep. 2019.

[33] D. Kim, M. Liu, S. Lee, and V. R. Kamat, "Remote proximity monitoring between mobile construction resources using camera-mounted UAVs", *Autom. Constr.*, no. 99, pp. 168-182, Mar. 2019.

[34] Q. Fang et al., "A deep learning-based method for detecting non-certified work on construction sites", *Adv. Eng. Inf.*, no. 35, pp. 56-68, Jan. 2018.

[35] J. Liu, W. Fang, P. E. D. Love, T. Hartmann, H. Luo, and L. Wang, "Detection and location of unsafe behaviour in digital images: A visual grounding approach", *Adv. Eng. Inf.*, no. 53, p. 101688, Aug. 2022.

[36] J. Li, X. Zhao, G. Zhou, and M. Zhang, "Standardized use inspection of workers' personal protective equipment based on deep learning", *Saf. Sci*., no. 150, p. 105689, Jun. 2022.

[37] Q. Fang et al., "Detecting non-hardhat-use by a deep learning method from far-field surveillance videos", *Autom. Constr.*, no. 85, pp. 1-9, Jan. 2018.

[38] J. Czarnowski, A. Dąbrowski, M. Maciaś, J. Główka, and J. Wrona, "Technology gaps in Human-Machine Interfaces for autonomous construction robots", *Autom. Constr.*, no. 94, pp. 179-190, Oct. 2018.

[39] B. Chu, K. Jung, M.-T. Lim, and D. Hong, "Robot-based construction automation: An application to steel beam assembly (Part I)", *Autom. Constr.*, no. 32, pp. 46-61, Jul. 2013.

[40] W. C. Stone, L. Pfeffer, and K. Furlani, "Automated Part Tracking on the Construction Job Site", *Robotics 2000*, pp. 96-103, Feb., 2000

[41] A. Kelm et al., "Mobile passive Radio Frequency Identification (RFID) portal for automated and rapid control of Personal Protective Equipment (PPE) on construction sites", *Autom. Constr.*, no. 36, pp. 38-52, Dec. 2013.

[42] T. Kong, W. Fang, P. E. D. Love, H. Luo, S. Xu, and H. Li, "Computer vision and long short-term memory: Learning to predict unsafe

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behaviour in construction", *Adv. Eng. Inf.*, no. 50, p. 101400, Oct. 2021.
[43] W. Fang et al., "A deep learning-based approach for mitigating falls from height with computer vision: Convolutional neural network", *Adv. Eng. Inf.*, no. 39, pp. 170-177, Jan. 2019.
[44] H. Kim, K. Kim, and H. Kim, "Vision-Based Object-Centric Safety Assessment Using Fuzzy Inference: Monitoring Struck-By Accidents with Moving Objects", *J. Comput. Civ. Eng.*, vol. 30, no. 4, p. 04015075, 2016.
[45] J. Cai, L. Yang, Y. Zhang, S. Li, and H. Cai, "Multitask Learning Method for Detecting the Visual Focus of Attention of Construction Workers", *J. Constr. Eng. Manage.*, vol. 147, no. 7, p. 04021063, Jul. 2021.
[46] X. Wang and Z. Zhu, "Vision–based framework for automatic interpretation of construction workers' hand gestures", *Autom. Constr.*, no. 130, p. 103872, Oct. 2021.
[47] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection", *Proc. IEEE Conf. Comput. Vis. Pattern Recog.*, pp. 779-788, June 2016.
[48] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137-1149, Jun. 2017.
[49] J. Redmon and A. Farhadi, "YOLO9000: Better, Faster, Stronger", *Proc. IEEE Conf. Comput. Vis. Pattern Recog*, pp. 6517-6525. July 2017.
[50] T. Liang, W. Pan, H. Bao, and F. Pan, "Vehicle wheel weld detection based on improved YOLO v4 algorithm," *Comput Opt*, 2022.
[51] Z. Xie, H. Liu, Z. Li, and Y. He, "A convolutional neural network based approach towards real-time hard hat detection", *IEEE Int. Conf. Prog. Inf. and Comput. (PIC)*, pp. 430-434, Dec. 2018.
[52] D. Wan, R. Lu, S. Shen, T. Xu, X. Lang, and Z. Ren, "Mixed local channel attention for object detection", *Eng. Appl. Artif. Intell.*, no. 123, p. 106442, Aug. 2023.
[53] G. Liu, Y. Hu, Z. Chen, J. Guo, and P. Ni, "Lightweight object detection algorithm for robots with improved YOLOv5", *Eng. Appl. Artif. Intell.*, no. 123, p. 106217, Aug. 2023.
[54] Q. Wang, M. Cheng, S. Huang, Z. Cai, J. Zhang, and H. Yuan, "A deep learning approach incorporating YOLO v5 and attention mechanisms for field real-time detection of the invasive weed Solanum rostratum Dunal seedlings", *Comput. Electron. Agric.*, no. 199, p. 107194, Aug. 2022.
[55] S. Chen, Y. Liao, F. Lin, and B. Huang, "An Improved Lightweight YOLOv5 Algorithm for Detecting Strawberry Diseases", *IEEE Access*, vol. 11, pp. 54080-54092, Aug. 2023.
[56] S. H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. D. Reid, and S. Savarese, "Generalized Intersection Over Union: A Metric and a Loss for Bounding Box Regression", *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 658-666.
[57] D. Chen and D. Miao, "Control Distance IoU and Control Distance IoU Loss Function for Better Bounding Box Regression", in arXiv, vol. abs/2103.11696, 2021.
[58] Y.-F. Zhang, W. Ren, Z. Zhang, Z. Jia, L. Wang, and T. Tan, "Focal and Efficient IOU Loss for Accurate Bounding Box Regression", *Neurocomputing*, vol. 506, pp. 146-157, 2021.
[59] J. He, S. M. Erfani, X. Ma, J. Bailey, Y. Chi, and X. Hua, "Alpha-IoU: A Family of Power Intersection over Union Losses for Bounding Box Regression", in arXiv, 2021.
[60] Z. Gevorgyan, "SIoU Loss: More Powerful Learning for Bounding Box Regression", in arXiv, vol. abs/2205.12740, 2022.
[61] X. Dong, S. Yan, and C. Duan, "A lightweight vehicles detection network model based on YOLOv5", *Eng. Appl. Artif. Intell.*, vol. 113, p. 104914, Aug. 2022.
[62] Ultralytics, "Yolov5, Available online: https://github.com/ultralytics/yolov5," 2021.
[63] F. Majeed et al., "Investigating the efficiency of deep learning based security system in a real-time environment using YOLOv5", *Sustainable Energy Technol. Assess.*, vol. 53, p. 102603, Oct. 2022.
[64] T. Sun, H. Xing, S. Cao, Y. Zhang, S. Fan, and P. Liu, "A novel detection method for hot spots of photovoltaic (PV) panels using improved anchors and prediction heads of YOLOv5 network", *Energy Rep.*, vol. 8, pp. 1219-1229, Nov. 2022.
[65] M. E. El-sayed, A. W. Youssef, O. M. Shehata, L. A. Shihata, and E. Azab, "Computer vision for package tracking on omnidirectional wheeled conveyor: Case study", *Eng. Appl. Artif. Intell.*, vol. 116, p. 105438, Nov. 2022.
[66] J. Liu, Z. Zhao, C. Lv, Y. Ding, H. Chang, and Q. Xie, "An image enhancement algorithm to improve road tunnel crack transfer detection", *Constr. Build. Mater.*, vol. 348, p. 128583, Sep. 19, 2022.
[67] R. Gai, M. Li, and N. Chen, "Cherry detection algorithm based on improved YOLOv5s network", IEEE 23rd Int Conf on High Performance Computing & Communications; 7th Int Conf on Data Science & Systems; 19th Int Conf on Smart City; 7th Int Conf on Dependability in Sensor, Cloud & Big Data Systems & Application (HPCC/DSS/SmartCity/DependSys), pp. 2097-2103, Dec. 2021.
[68] K. Wang, J. H. Liew, Y. Zou, D. Zhou, and J. Feng, "PANet: Few-Shot Image Semantic Segmentation With Prototype Alignment", *Proc. IEEE Int. Conf. Comput. Vis.*, pp. 9196-9205, Oct. 2019.
[69] T.-Y. Lin, P. Dollár, R. B. Girshick, K. He, B. Hariharan, and S. J. Belongie, "Feature Pyramid Networks for Object Detection", *Proc. IEEE Conf. Comput. Vis. Pattern Recog.*, pp. 936-944, 2017.
[70] K. He, X. Zhang, S. Ren, and J. Sun, "Spatial Pyramid Pooling in Deep Convolutional Networks for Visual Recognition", *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1904-1916, 2015.
[71] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient Channel Attention for Deep Convolutional Neural Networks", *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 11531-11539, Jun. 2020.
[72] Q. Hou, D. Zhou, and J. Feng, "Coordinate Attention for Efficient Mobile Network Design", *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 13708-13717, Jun. 2021.
[73] A. Paszke et al., "PyTorch: An Imperative Style, High-Performance Deep Learning Library", in NeurIPS, 2019.
[74] W. Fang, L. Ding, H. Luo, and P. E. D. Love, "Falls from heights: A computer vision-based approach for safety harness detection", *Autom. Constr.*, vol. 91, pp. 53-61, Jul. 2018.
[75] W. Fang, L. Ding, B. Zhong, P. E. D. Love, and H. Luo, "Automated detection of workers and heavy equipment on construction sites: A convolutional neural network approach", *Adv. Eng. Inf.*, vol. 37, pp. 139-149, Aug. 2018.
[76] H. Chen, L. Hou, G. Zhang, and S. Wu, "Using Context-Guided data Augmentation, lightweight CNN, and proximity detection techniques to improve site safety monitoring under occlusion conditions", *Saf. Sci.*, vol. 158, p. 105958, Feb. 2023.
[77] Z. Situ et al., "A transfer learning-based YOLO network for sewer defect detection in comparison to classic object detection methods", *Dev. Built Environ.*, vol. 15, p. 100191, Oct. 2023.
[78] A. M. Roy and J. Bhaduri, "DenseSPH-YOLOv5: An automated damage detection model based on DenseNet and Swin-Transformer prediction head-enabled YOLOv5 with attention mechanism", *Adv. Eng. Inf.*, vol. 56, p. 102007, Apr. 2023.

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Liu Yipeng, born in Wuhan, Hubei, China, embarked on his academic journey at Wuhan University of Technology in 2013, majoring in Materials Science and Engineering. He earned his bachelor's degree from 2013 to 2017, followed by a master's degree from the School of Civil Engineering and Architecture at Wuhan University of Technology from 2018 to 2021. In 2021, he commenced his doctoral studies in Civil Engineering at Wuhan University of Technology.

During his doctoral program, he demonstrated exceptional research prowess, contributing significantly to the scientific community. Over a span of two years, he authored three impactful papers in high-tier SCI journals. Notably, two of these publications were in JCR Q1 journals, and one in a JCR Q2 journal. His research primarily revolves around carbon emission computation in the construction industry and computer vision-based safety management in construction.

In the realm of carbon emission computation, Liu's work has contributed valuable insights to the assessment and reduction of environmental impact within the construction sector. His expertise extends to utilizing cutting-edge technologies like computer vision for enhancing safety protocols in construction projects. By leveraging innovative approaches, he aims to elevate safety standards and reduce occupational risks in the field of construction engineering.

Apart from his academic pursuits, Liu Yipeng has actively engaged in various aspects of university life, demonstrating leadership and commitment. As a driven and dedicated individual, he is poised to continue making meaningful contributions to the fields of materials science, construction engineering, and safety management.



Junwu Wang is a dedicated researcher and safety advocate with a strong commitment to improving workplace safety and reducing occupational hazards. Holding a Ph.D. in Civil Engineering. From Wuhan University of Technology, Wang has focused their career on exploring innovative solutions to complex safety challenges.

From 2002.07 to now, he has been awarded as Professor of Wuhan University of Technology, Bo-director, Director of Engineering Management Department, Director of Hubei Civil Engineering Society, Executive Director of Wuhan Real Estate Appraisers Association, Executive Director of Hubei Construction Costing Association, etc. He was recognized as Associate Professor in 2000 and Professor in 2005 in Wuhan University of Technology. He was recognized as associate professor of Wuhan University of Technology in 2000 and professor of Wuhan University of Technology in 2005. For many years, he has been engaged in scientific research and teaching of civil engineering construction and management. In recent years, he has presided over and participated in more than 30 scientific research projects of National Natural Science Foundation of China, provincial and ministerial level, and other scientific research projects, published more than 80 academic papers in core academic journals at home and abroad and in international academic conferences, published one monograph, and graduated more than 100 double certificates of PhDs and masters. He has made creative researches in the fields of civil engineering project investment risk decision-making, sustainable development and evaluation, theory and application of project financing mode.

Throughout their career, he has collaborated with industry experts, safety professionals, and government agencies to develop practical strategies for enhancing safety measures across diverse work environments. Their research often emphasizes the application of cutting-edge technologies, including computer vision, machine learning, and data analytics, to improve safety outcomes.

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